Team members:

Today we will investigate why different random samples can lead to different estimates, even when everyone follows the same sampling procedure.

Part A:

1. Suppose we want to estimate the mean number of hours studied over a weekend by the students in the Data Science Club. We randomly sample 6 of the students and record their study times: 2, 3, 4, 4, 5, and 7 hours. Identify the:
    a. Population:

    b. Sample:

    c. Variable of interest (and type):

    d. Population parameter:

    e. Sample statistic:

    f. Sample size:

2. Suppose another group separately randomly selects 6 students from the Data Science Club. Which of the components from question 1 (a-f) must stay the same and which could change?

    Same: Change:

3. A student says:
    "If we select students randomly, our sample mean should equal the population mean."
    Do you agree? Explain what random sampling does—and whether it guarantees an exact estimate.

Part B:

1. Now suppose we want to estimate the proportion of Data Science Club students who plan to start a job the summer after graduating (versus other plans like graduate school, traveling abroad, etc.). We randomly sample 10 of the students and record whether or not they plan to start a job: yes, yes, no, no, no, yes, no, yes, yes, yes. Identify the:
    a. Population:

    b. Sample:

    c. Variable of interest (and type):

    d. Population parameter:

    e. Sample statistic:

    f. Sample size:

2. Suppose another group randomly selects 10 students from the Data Science Club. Which of the components from question 1 (a-f) must stay the same and which could change?

Same: Change:

Part C:

Now you will do a hands on exercise to simulate sampling members of the Data Science Club. Each bag contains 30 beads representing the 30 members of the club, the population. Each bead represents one student.

_____ beads represent students who plan to start a job.

_____ beads represent students who have other plans.

In this example we have access to the population. What is the true proportion of Data Science Club members that plan to start a job?

1. Mix the beads in your bag. Without looking, select **10 beads**. Keep the selected beads out of the bag until you have all 10 so that a student cannot be selected twice in the same sample.
2. Count how many of the selected students plan to start a job. Calculate the sample proportion and record the results of your sample in the table.
3. Return all 10 beads to the bag and mix thoroughly. Repeat the procedure twice more and record your results in the table.

Sample	Number who plan to start a job	Sample proportion
1		
2		
3		

4. Compare your three sample proportions.
   a. Are they all the same? If not, how different are they?

   b. What was your sample size for each sample?

   c. Explain why the sample proportions could differ even though each sample came from the same population.

Part D:

Dr. Medina will collect the sample proportions found by each pair of students in this class and make a histogram.

1. How many sample proportions are represented in the histogram? How many students were included in **each sample**?

2. Describe the shape of the histogram.

3. What value is the histogram centered around? How does this compare with the population proportion?

Part E:

Now we will repeat the same process as parts C and D, but with a sample size of 20.

1. Draw a sample of size 20 and calculate your sample proportion of students planning to start a job the summer after graduating.

2. Report your sample proportion to Dr. Medina to help create a histogram for samples of size 20.
3. Compare this histogram to the histogram of sample proportions from samples of size 10. How do they differ?

The variation in a sample statistic from one random sample to another is called **sampling variability**.

4. Explain sampling variability in your own words using this activity. Identify what stayed the same and what could change each time we selected a sample.